Olist Commerce360

An End-to-End E-Commerce Data Warehouse and Business Intelligence Platform

P1_05 - Grain Definition Document

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1. Purpose

The grain states exactly what one row of a table represents. It is declared before the measures are chosen, and every measure and foreign key in the table must be true at that grain. The Olist source records orders, items, payments and reviews at different grains. A clear grain is therefore the main protection against double counting.

2. Fact table grains

fact_order_items

Grain statement	**One row per product line within a customer order (order_id + order_item_id). Each row is one unit of one product sold by one seller.**
Fact type	Transaction fact
Uniqueness key	order_id + order_item_id (verified unique in source: 0 duplicates)
Source rows / expected rows	112,650
Row source	olist_order_items_dataset.csv joined to olist_orders_dataset.csv and olist_customers_dataset.csv for keys
Measures valid at this grain	item_price, freight_value, item_gross_value, item_quantity
Dimensions at this grain	dim_customer, dim_date, dim_order_status, dim_product, dim_seller

fact_orders

Grain statement	**One row per customer order (order_id), updated as the order passes purchase, approval, carrier hand-off and delivery milestones.**
Fact type	Accumulating snapshot fact
Uniqueness key	order_id (verified unique in source: 0 duplicates)
Source rows / expected rows	99,441
Row source	olist_orders_dataset.csv + aggregates of order_items, order_payments and order_reviews per order
Measures valid at this grain	item_count, order_item_value, order_freight_value, order_payment_value, payment_count, days_purchase_to_approval, days_approval_to_carrier, days_carrier_to_customer, days_purchase_to_delivery, days_promised, delivery_delay_days, is_delivered, is_late, customer_order_number, is_repeat_purchase, review_count, avg_review_score
Dimensions at this grain	dim_customer, dim_date, dim_order_status

fact_payments

Grain statement	**One row per payment record of an order (order_id + payment_sequential). An order paid with a card and two vouchers has three rows.**
Fact type	Transaction fact
Uniqueness key	order_id + payment_sequential (verified unique in source: 0 duplicates)
Source rows / expected rows	103,886
Row source	olist_order_payments_dataset.csv joined to olist_orders_dataset.csv for customer, status and date
Measures valid at this grain	payment_value, payment_installments
Dimensions at this grain	dim_customer, dim_date, dim_order_status, dim_payment_type

fact_reviews

Grain statement	**One row per review submitted for an order (review_id + order_id). The same review_id can cover more than one order, and an order can receive more than one review.**
Fact type	Transaction fact
Uniqueness key	review_id + order_id (verified unique in source: 0 duplicates)
Source rows / expected rows	99,224
Row source	olist_order_reviews_dataset.csv joined to olist_orders_dataset.csv for customer
Measures valid at this grain	review_score, response_hours, has_comment_title, has_comment_message
Dimensions at this grain	dim_customer, dim_date

Inclusion rules

- fact_orders contains every order (99,441), including canceled, unavailable and undelivered orders. Of these, 775 orders have no items, so item_count = 0 and the value measures are 0. Status filters are applied at query time through dim_order_status, not by deleting rows.
- fact_order_items contains every item line (112,650), including lines of cancelled orders. Revenue KPIs exclude them through status_group.
- fact_payments contains every payment record (103,886), including 9 records with value 0 and 3 with type 'not_defined'.
- fact_reviews contains every review/order pair (99,224).

3. Dimension grains

Dimension	One row represents	Natural key	Expected rows
dim_date	One row per calendar day from 2016-01-01 to 2018-12-31, plus one 'Unknown / not occurred' row (date_key = -1).	full_date	1,096 + 1 unknown
dim_customer	One row per real customer (customer_unique_id), plus one 'Unknown' row (customer_key = -1).	customer_unique_id	96,096 + 1 unknown
dim_product	One row per product (product_id), plus one 'Unknown' row (product_key = -1).	product_id	32,951 + 1 unknown
dim_seller	One row per seller (seller_id), plus one 'Unknown' row (seller_key = -1).	seller_id	3,095 + 1 unknown
dim_payment_type	One row per payment method code, plus one 'Unknown' row (payment_type_key = -1).	payment_type_code	5 + 1 unknown
dim_order_status	One row per order status code, plus one 'Unknown' row (order_status_key = -1).	order_status_code	8 + 1 unknown

4. Relationship between fact grains (verified in source)

Child process	Orders with rows	Orders without rows	Orders with >1 row	Max per order	Mean per order
Order items	98,666	775	9,803	21	1.142
Payments	99,440	1	2,961	29	1.045
Reviews	98,673	768	547	3	1.006

Order -> items, order -> payments and order -> reviews are each one-to-many with different cardinalities. Items and payments are not related to each other below order level: a payment is not allocated to a particular item.

5. Join and aggregation rules

Evidence from the source: joining order_items to order_payments on order_id produces 117,601 rows. SUM(price) rises from BRL 13,591,643.70 to 14,209,115.34, and SUM(payment_value) rises from BRL 16,008,872.12 to 20,308,134.71.

Rule	Statement
G-01	Never join two fact tables row-to-row on order_id. To compare processes, aggregate each fact separately to a shared dimension level (month, state, customer) and then combine the results (drill-across).
G-02	Count orders from fact_orders (COUNT(*)), or COUNT(DISTINCT order_id) on other facts. Never count item or payment rows as orders.
G-03	Item sales revenue comes only from fact_order_items.item_price, and amount paid only from fact_payments.payment_value. fact_orders holds per-order totals of both for order-level analysis.
G-04	Per-order aggregates in fact_orders (item_count, order_item_value, order_freight_value, order_payment_value, payment_count, review_count, avg_review_score) are computed by grouping each child table by order_id before joining to orders.
G-05	Non-additive measures (scores, durations, instalments, customer_order_number) are averaged or used as filters, never summed.
G-06	Review analysis against delivery is done at order grain in fact_orders (avg_review_score with is_late), so no fact-to-fact join is needed.
G-07	Product and seller are only valid at item grain. Questions that combine product or seller with delivery or review measures need an explicit allocation rule and are outside the Phase 1 scope (see P1_03 section 5).
G-08	In Power BI, import each fact and dimension as a separate table with single-direction dimension (1) -> fact (*) relationships. Do not create relationships between fact tables.

6. Reconciliation targets for Phase 2

Check	Expected value (from source)
fact_order_items row count	112,650
fact_orders row count	99,441
fact_payments row count	103,886
fact_reviews row count	99,224
dim_customer rows (excl. unknown)	96,096
dim_product rows (excl. unknown)	32,951
dim_seller rows (excl. unknown)	3,095
SUM(fact_order_items.item_price) = SUM(fact_orders.order_item_value)	13,591,643.70
SUM(fact_order_items.freight_value) = SUM(fact_orders.order_freight_value)	2,251,909.54
SUM(fact_payments.payment_value) = SUM(fact_orders.order_payment_value)	16,008,872.12
Fact rows with an FK not found in its dimension	0